

1) Series
a) Arithmetic progressions

Example 5

The 1st term of an arithmetic progression is -6 and the common difference is 5 .
Find the 4th term and the n th term.

$$a = -6 \quad d = 5$$

$$\text{4th term} = a + 3d$$

$$= -6 + (3)5 = 9$$

Substitute $a = -6$ and $d = 5$

$$n\text{th term} = a + (n - 1)d$$

$$= -6 + (n - 1)5 = 5n - 11$$

Substitute $a = -6$ and $d = 5$

Example 6

The n th term of an arithmetic progression is $7n - 2$. Find the 1st term and the common difference, d .

The 1st term has $n = 1$

$$u_1 = 7(1) - 2 = 5$$

Substitute $n = 1$

The 2nd term has $n = 2$

$$u_2 = 7(2) - 2 = 12$$

Substitute $n = 2$

$d = 2\text{nd term} - 1\text{st term}$

$$d = 12 - 5 = 7$$

Example 7

The 5th term of an arithmetic progression is 21 and the sum of the first six terms is 90 .
Find the 18th term.

$$\text{5th term} = a + 4d = 21$$

$$S_6 = \frac{6}{2} [2a + (6 - 1)d] = 90$$

$$\text{Thus, } 6a + 15d = 90$$

Solving $a + 4d = 21$ and $6a + 15d = 90$ simultaneously gives $a = 5$ and $d = 4$.

$$\text{18th term} = a + 17d$$

$$= 5 + 17(4) = 73$$

Example 8

The sum of the first n terms of an arithmetic progression is $n^2 + 5n$.

a) Find the n th term. b) Write down the first four terms.

$$\text{a) } S_1 = 1^2 + 5(1) = 6 = \text{first term}$$

$$S_2 = 2^2 + 5(2) = 14 = \text{first term} + \text{second term}$$

$$\text{Second term} = S_2 - S_1 = 14 - 6 = 8$$

$$a = 6, d = 2,$$

$$n\text{th term} = 6 + (n - 1)2 = 2n + 4$$

Or

$$\text{The } n\text{th term} = S_n - S_{n-1} = n^2 + 5n - [(n - 1)^2 + 5(n - 1)]$$

$$= n^2 + 5n - [n^2 - 2n + 1 + 5n - 5]$$

$$= 2n + 4$$

Substitute $n = 1$ into $n^2 + 5n$.

Substitute $n = 2$ into $n^2 + 5n$.

For S_{n-1} put $n - 1$ instead of n .

$$\text{b) First four terms} = 2(1) + 4, 2(2) + 4, 2(3) + 4, 2(4) + 4 \\ = 6, 8, 10, 12$$

Note: It is useful to remember that the n th term of any progression is $S_n - S_{n-1}$.

b) Geometric progression

Example 9

The 3rd term of a geometric progression is 9 and the common ratio is -3 .
Find the 6th term and the n th term.

$$\begin{array}{llll}
 \text{3rd term} & = ar^2 & = 9 & a \times (-3)^2 = 9 \quad a = \frac{9}{9} = 1 \quad \leftarrow \text{Substitute } r = -3. \\
 \text{6th term} & = ar^5 & = 1(-3)^5 & = -243 \\
 \text{\textit{n}th term} & = ar^{n-1} & = (1)(-3)^{n-1} & = (-3)^{n-1}
 \end{array}$$

Example 10

The n th term of a geometric progression is $10\left(\frac{1}{2}\right)^{n-1}$. Find the 1st term and the common ratio.

$$\begin{array}{ll}
 \text{1st term} = u_1 = 10\left(\frac{1}{2}\right)^0 = 10 & \leftarrow \text{Substitute } n = 1. \\
 a = 10 & \\
 \text{2nd term} = u_2 = 10\left(\frac{1}{2}\right)^1 = 5 & \leftarrow \text{Substitute } n = 2. \\
 r = \text{2nd term} \div \text{1st term} = 5 \div 10 & \\
 r = \frac{1}{2} &
 \end{array}$$

The **sum of a geometric progression** can be written as $S_n = \frac{a(1-r^n)}{1-r}$, $r \neq 1$.

We use this formula when $-1 < r < 1$.

When we do not have $-1 < r < 1$, we use the formula $S_n = \frac{a(r^n-1)}{r-1}$.

Proof:

$$\begin{array}{ll}
 S_n = a + ar + ar^2 + \dots + ar^{n-3} + ar^{n-2} + ar^{n-1} & \\
 rS_n = ar + ar^2 + \dots + ar^{n-2} + ar^{n-1} + ar^n & \leftarrow \text{Multiply each term by } r. \\
 S_n - rS_n = a - ar^n & \leftarrow \text{Subtract the 2nd line from the 1st line.} \\
 S_n(1-r) = a(1-r^n) & \leftarrow \text{Factorise each side.} \\
 S_n = \frac{a(1-r^n)}{1-r} & \leftarrow \text{Use when } |r| < 1. \\
 \text{If we multiply by } \frac{-1}{-1} \text{ we get } S_n = \frac{a(r^n-1)}{r-1} & \leftarrow \text{Use when } |r| > 1.
 \end{array}$$

Note: This is not defined when $r = 1$.

Example 11

The 3rd term of a geometric progression is 16 and the 6th term is -128 .
Find the 1st term and the sum of the first seven terms.

$$\begin{array}{ll}
 ar^2 = 16 & ar^5 = -128 \\
 \frac{ar^5}{ar^2} = \frac{-128}{16} & \leftarrow \text{With geometric progressions we divide the equations to solve simultaneously.} \\
 r^3 = -8 & r = -2 \\
 ar^2 = 16 & a(-2)^2 = 16 \quad a = 16 \div 4 = 4 \\
 S_7 = \frac{a(r^7-1)}{r-1} = \frac{4((-2)^7-1)}{-2-1} & \leftarrow \text{Substitute } a = 4, r = -2 \text{ and } n = 6. \\
 = \frac{4(-128-1)}{-3} = 172 &
 \end{array}$$

Example 12

Mabintou deposits \$100 in a bank account at the start of each year.
She earns 4% compound interest.

- a) Work out how much money she has in her bank account at the end of the 3rd year.
- b) Work out how much money she has in her account at the start of the 30th year after she has made her annual deposit.

- a) At the end of the third year the amount in her account is
- $$\begin{aligned} & \$100(1.04) + \$100(1.04)^2 + \$100(1.04)^3 \\ & = \$104 + \$108.16 + \$112.4864 = \$324.65 \end{aligned}$$

- b) At the start of the 30th year, in \$, she has
- $$100 + 100(1.04) + 100(1.04)^2 + \dots + 100(1.04)^{29}$$
- $$a = 100 \quad r = 1.04 \quad n = 30$$

$$S_{30} = \frac{100(1.04^{30} - 1)}{1.04 - 1} = \$5608.49 \text{ (to the nearest cent)}$$

Example 13

An arithmetic progression has first term a and common difference d .

The 1st, 3rd and 13th terms of the arithmetic progression form the first three terms of a geometric progression with common ratio r .

- a) Express d in terms of a , and find the value of r .
- b) The 4th term in the geometric progression also appears in the arithmetic progression. Determine which term it is.

- a) $a, a + 2d, a + 12d$ are the first three terms in the geometric progression.

$$\text{So } (a + 2d)^2 = a(a + 12d)$$

$$a^2 + 4ad + 4d^2 = a^2 + 12ad$$

$$4d^2 = 8ad$$

$$d = 2a$$

$$\text{So } a + 2d = a + 2 \times 2a = 5a$$

The 2nd term of the geometric progression is $5a$, so $r = 5$.

- b) The 4th term in the geometric progression is $ar^3 = 125a$.

If this is the n th term of the arithmetic progression with first term a and common difference $2a$:

$$125a = a + (n - 1)(2a) \Rightarrow n = 63$$

So the 4th term in the geometric progression is the 63rd term in the arithmetic progression.

These are the 1st, 3rd and 13th terms of an arithmetic progression.

If a, b and c are consecutive terms in the geometric progression, then $\frac{b}{a} = \frac{c}{b}$, so $b^2 = ac$.

Expand the bracket and simplify the arithmetic progression.

Substitute $d = 2a$.

2nd term $\div$ 1st term $= 5a \div a$

Substitute $r = 5$.

c) Infinite geometric progressions

Example 14

The n th term of a geometric progression is $81 \left(-\frac{1}{3}\right)^n$. Find the sum to infinity.

$$\begin{aligned} \text{1st term: } u_1 &= 81 \left(-\frac{1}{3}\right)^1 = -27 \\ a &= -27 \end{aligned}$$

← Substitute $n = 1$.

$$\text{2nd term: } u_2 = 81 \left(-\frac{1}{3}\right)^2 = 9$$

← Substitute $n = 2$.

$$r = \frac{u_2}{u_1} = \frac{9}{-27} = -\frac{1}{3}$$

$$S_\infty = \frac{a}{1-r} = -\frac{27}{1 - \left(-\frac{1}{3}\right)} = -20\frac{1}{4}$$

← Substitute $a = -27$ and $r = -\frac{1}{3}$.

Example 15

a) Write the recurring decimal $0.4\dot{5}$ as the sum of a geometric progression.

b) Use your answer to part (a) to show that $0.4\dot{5}$ can be written as $\frac{41}{90}$.

$$\text{a) } 0.4\dot{5} = 0.4 + 0.0\dot{5} = 0.4 + \frac{5}{100} + \frac{5}{1000} + \frac{5}{10000} + \dots$$

$$\text{b) Consider } \frac{5}{100} + \frac{5}{1000} + \frac{5}{10000} + \dots \quad a = \frac{5}{100} \quad r = \frac{1}{10}$$

$$S_\infty = \frac{a}{1-r} = \frac{\frac{5}{100}}{1 - \frac{1}{10}} = \frac{5}{100} \times \frac{10}{9} = \frac{5}{90}$$

$$\text{Thus } 0.4\dot{5} = 0.4 + 0.0\dot{5} = \frac{2}{5} + \frac{5}{90} = \frac{41}{90}$$

Summary

Terms of a sequence

- u_n can be used to describe any term in a sequence. u_1 is the first term.
- The term after u_n is u_{n+1} .
- The term before u_n is u_{n-1} .

Finite and infinite series

- A series is **finite** if it has a finite number of terms, e.g. $5 + 9 + 13 + 17 + 21$.
- A series is **infinite** if it continues indefinitely, e.g. $5 + 9 + 13 + 17 + 21 + \dots$
- The **general term** of the series $5 + 9 + 13 + 17 + 21 + \dots$ is $4r + 1$.
- The finite series $5 + 9 + 13 + 17 + 21 + 25$ can be written as $\sum_{r=1}^6 (4r + 1)$.
- The infinite series $5 + 9 + 13 + 17 + 21 + \dots$ can be written as $\sum_{r=1}^{\infty} (4r + 1)$.

Arithmetic progressions

- $a = 1\text{st term}$ ($= u_1$)
- $d = \text{common difference}$ ($= u_2 - u_1 = u_3 - u_2 = u_4 - u_3$ etc.)
- $l = \text{last term in a finite series}$
- An arithmetic progression with n terms can be written as $a, (a + d), (a + 2d), (a + 3d), \dots, \dots, [a + (n - 1)d]$.
- The **n th term** of an arithmetic progression can be written as $[a + (n - 1)d]$.
- The **n th term** is $l = a + (n - 1)d$.
- We can use this formula to write down an expression for any term.
e.g. 8th term $= a + 7d$ e.g. 20th term $= a + 19d$.

Geometric progressions

- A **geometric progression** is a series that has a **common ratio** between each term.
- $a = 1\text{st term}$ ($= u_1$)
- $r = \text{common ratio}$ ($= u_2 \div u_1 = u_3 \div u_2 = u_4 \div u_3$ etc.)
- $l = \text{last term in a finite series}$
- A geometric progression with n terms can be written as $a, ar, ar^2, ar^3, \dots, \dots, ar^{n-1}$.
- The **n th term** of a geometric progression can be written as ar^{n-1} .
- The **n th term** is $l = ar^{n-1}$.
- We can use this formula to write down an expression for any term.
e.g. 8th term $= ar^7$ e.g. 20th term $= ar^{19}$
- The **sum of a geometric progression** can be written as $S_n = \frac{a(1 - r^n)}{1 - r}$.
- The sum to infinity of a geometric progression is $S_{\infty} = \frac{a}{1 - r}$ provided $|r| < 1$.